

URA743: AUTOMOTIVE MECHATRONICS SYSTEMS

L	T	P	Cr
3	0	0	3.0

Course Objective: To prepare the students to critically evaluate the challenges and identify the role of electronics and software systems in a modern automobile. Students are taught basic automotive mechatronic systems, underlying principles of construction and working, limitations of the conventional systems, the needs for electronic controls to improve the performance. Also, they are motivated to explore potential new functions and applications by studying the physical systems.

Syllabus

Introduction to Automotive Electronics: Vehicle system architecture, Functional Structure, Function modules of an electronic system, Electronic Control Unit (ECU), Control mechanisms, Cross-system functions, Requirements for Bus Systems, CAN Bus, LIN Bus, Network Topologies.

Automotive Sensors and Measurements:

Position sensors, Speed and rpm sensors, Acceleration sensors, Pressure sensors, Force and torque sensors, Flowmeters, Gas sensors and concentration sensors, Temperature sensors, Imaging sensors

Electronic Control of Combustion: Electronic combustion control; control of exhaust gas recirculation; electronic control of valve timing; distributorless ignition systems (DIS); fuel metering and control; electronic fuel injection: single point, multi point and direct injection.

Advanced Automotive Lighting: Computer controlled headlight systems; automatic on/off with time delay; automatic headlight dimming; headlight levelling; adaptive headlights; adaptive Brake Lights; instrument panel dimming; lamp outage indicators; LED lamps; cornering lights.

Driver Assistance System: Digital instrument cluster; travel information system; head-updisplay; night vision system; global positioning navigation system; lane change warning system; warning programs; traffic management system; hand's free communication and operation.

Automatic Transmission and Driveline: Drive by wire system; electronic shift transmission; direct shift gearbox; S-Tronic gearbox; paddle shift control; constantly

variable transmission; cruise control; limited slip differential; differential lock; hill ascent function.

Active and Passive Safety Systems: Introduction; primary restraint system; secondary restraint system; passive seat belt systems; air bag systems; seat belt pre-tensioners; inflatable knee blockers; occupant classification systems; anti-whiplash headrest restraint system; antilock braking system.

On-Board Diagnosis (OBD): Vehicle management system; requirement of OBD-II; fault diagnosis; fault codes; vehicle fault scanners; vehicle oscilloscope.

Course Learning Outcomes (CLOs)

The students will be able to:

1. correlate the function of electronic systems in modern automobiles.
2. evaluate the use of electronics to automate mechanical controls for improved performance, safety, comfort and related issues.
3. specify the intervention of electronics devices in saving fuel by controlling combustion process and pollutants.

Text Books

1. Konrad Reif Ed. , Automotive Mechatronic: Automotive Networking, Driving Stability Systems, Electronics, Bosch Professional Automotive Information, Springer, (2015)
2. Hiller, V. A. W., Fundamentals of Automotive Electronics, 6th Edition, Oxford University Press, UK (2014).
3. Hiller, V. A. W., Fundamentals of Motor Vehicle Technology, 6th Edition, Oxford university Press, UK (2012).

Reference Books

1. Hollembeak, B., Today's Technician: Advanced Automotive Electronic Systems, 6th Edition, Delmar Cengage Learning, USA (2014).
2. Heisler, H., Advanced Vehicle Technology, 2nd Edition, Butterworth-Heinemann (2002)
3. Pike , J. A., Automotive Safety, 1st Edition, SAE International (2009).